

Sathya Baarathi Shanthi Ravichandran, Ph.D

The Jackson Laboratory for Genomic Medicine | 10 Discovery Drive, Farmington, CT, USA
sathya.ravi@jax.org | sathyabaarathi@gmail.com | <https://sathyabaarathisr.github.io/> | +1 (860) 964-7654

PROFILE SUMMARY

Systems immunologist with **19 peer-reviewed publications**, including work in *Nature Immunology* (IF 27.6), *EBioMedicine* (The Lancet Group; IF 10.8), and *Seminars in Immunology* (IF 7.8), and **two patents as lead inventor**. Research published in *Nature Immunology* has been **independently adopted by clinical trial investigators** and cited in *Nature Reviews Cancer*, *JCI*, and *NPJ Vaccines*; diagnostic biomarkers reported in *EBioMedicine* have been **validated across cohorts in 19 countries**. Recipient of the **Patterson Trust Award (\$200,000) and the Brooks Scholar Fellowship**. Serves as **Handling Editor** (Frontiers journals) and invited **peer reviewer** for publishers including Springer Nature, BioMed Central (BMC), Frontiers, and MDPI.

POSITIONS

Postdoctoral Associate | Systems immunologist Oct 2021 – Present

The Ucar Lab, The Jackson Laboratory for Genomic Medicine, Farmington, CT

Research Associate Sep 2020 – Sep 2021

Chandra Lab, Indian Institute of Science, Bangalore, India

Visiting Student Oct – Dec 2016

Department of Applied Mathematics, University of Leeds, UK

Bioinformatics Scientist Jul 2010 – Jan 2012

Vision Research Foundation, Sankara Nethralaya, India

EDUCATION

Ph.D. in Science Aug 2013 – Aug 2020

Indian Institute of Science, Bangalore, India

B.Tech. in Bioinformatics Jun 2006 – May 2010

SASTRA University, Thanjavur, India

AWARDS & HONORS

- Robert E. Leet & Clara Guthrie Patterson Trust Mentored Research Award (2026–2028) - Health Resources in Action, Boston, MA - \$200,000 independent competitive funding
- Brooks Scholar Postdoctoral Fellowship (2023–2024) - The Jackson Laboratory Cancer Center
- AAI Trainee Abstract Award (2023), American Association of Immunologists, IMMUNOLOGY 2023, Washington, DC
- Best Poster Award, 32nd Annual Short Course on Experimental Models of Human Cancer, The Jackson Laboratory, Bar Harbor, ME (2023)
- Best Poster Award, 2nd Biennial Meeting, Sex Differences in the Immune System, Trinity College Dublin & Whitehead Institute
- Global Health Travel Award, Keystone Symposia, Translational Systems Immunology, Snowbird, UT (2018)
- Global Travel Award, Boehringer Ingelheim Fonds (2016)

- Dean's List for Academic Excellence, SASTRA University (2008–2010)

PUBLICATIONS

Google Scholar: <https://scholar.google.com/citations?user=uhoSPP4AAAAJ>

† First author has provided written testimony confirming equal and independent scientific contribution.
(Total citations=582, h index =10)

Peer-Reviewed Articles

1. **Ravichandran S**, Erra-Diaz F, Karakaslar OE, Marches R, Kenyon-Pesce L, Rossi R, Chaussabel D, Nehar-Belaid D, LaFon DC, Pascual V, Palucka K, Ucar D, Banchereau J, Kuchel GA. Distinct baseline immune characteristics associated with responses to conjugated and unconjugated pneumococcal polysaccharide vaccines in older adults. *Nature Immunology*. 2024; 25(1):1-4. **(Impact Factor: 27.6)**
2. Nehar-Belaid D*, Sokolowski M*, **Ravichandran S***, Banchereau J, Chaussabel D, Ucar D. Baseline immune states (BIS) associated with vaccine responsiveness and factors that shape the BIS. *Seminars in Immunology*. 2022; 70:101842. **(Impact Factor: 7.8)** - Co-first author (* equal contribution)
3. Thakur C, Tripathi A, **Ravichandran S**, Shivananjaiah A, Chakraborty A, Varadappa S, Chikkavenkatappa N, Nagarajan D, Lakshminarasimhaiah S, Singh A, Chandra N. A new blood-based RNA signature (R9), for monitoring effectiveness of tuberculosis treatment in a South Indian longitudinal cohort. *iScience*. 2022; 25(2):103745. **(Impact Factor: 4.1)**
4. **Ravichandran S**, Banerjee U, Devi G, Kandukuru R, Thakur C, Chakravorty D, Narayanaswamy B, Singh A, Chandra N. VB10, a new blood biomarker for differential diagnosis and recovery monitoring of acute viral and bacterial infections. *EBioMedicine*. 2021; 67:103352. **(Impact Factor: 10.8)**
5. Nagarajan D, Roy N, Kulkarni O, Nanajkar N, Datey A, **Ravichandran S**, Thakur C, Aprameya IV, Sarma SP, Chakravorty D. Omega76: A designed antimicrobial peptide to combat carbapenem- and tigecycline-resistant *Acinetobacter baumannii*. *Science Advances*. 2019; 5(7):eaax1946. **(Impact Factor: 12.5)** †
6. **Ravichandran S**, Chandra N. Interrogation of genome-wide networks in biology: comparison of knowledge-based and statistical methods. *Int. J. Advances in Engineering Sciences and Applied Mathematics*. 2019; 11(2):119-137. **(Impact Factor: 0.8)**
7. Nagarajan D, Nagarajan T, Roy N, Kulkarni O, **Ravichandran S**, Mishra M, Chakravorty D, Chandra N. Computational antimicrobial peptide design and evaluation against multidrug-resistant clinical isolates of bacteria. *Journal of Biological Chemistry*. 2018; 293(10):3492-3509. **(Impact Factor: 3.9)** †
8. Sreenivasan S, **Ravichandran S**, Vetrivel U, Krishnakumar S. Modulation of multidrug resistance 1 expression and function in retinoblastoma cells by curcumin. *Journal of Pharmacology and Pharmacotherapeutics*. 2013; 4(2):103-109. **(Impact Factor: 0.8)**
9. Vetrivel U, **Ravichandran S**, Kuppan K, Mohanlal J, Das UN, Narayanasamy A. Agonistic effect of polyunsaturated fatty acids (PUFAs) on brain-derived neurotrophic factor (BDNF). *Lipids in Health and Disease*. 2012; 11(1):1-8. **(Impact Factor: 4.2)**
10. Harishankar A, Jambulingam M, Gowrishankar R, Vetrivel U, **Ravichandran S** et al. Phylogenetic comparison of HSV1 isolates: exonic US4, US7, UL44 regions. *Virology Journal*. 2012; 9(1):1-8. **(Impact Factor: 4.0)**
11. Vinita K, Sripriya S, Prathiba K, Vaitheeswaran K, **Ravichandran S** et al. ICAM-1 K469E polymorphism and diabetic retinopathy. *BMJ Open*. 2012; 2(4):e001036. **(Impact Factor: 2.4)**

12. Sreenivasan S, **Ravichandran S**, Umashankar V, Subramanian K. Inhibitory effects of curcumin on MRP1 in retinoblastoma cells. *Bioinformation*. 2012; 8(1):13. **(Impact Factor: 1.9)**
13. Jambulingam M, Vetrivel U, **Ravichandran S** et al. Functional characterization of UL54 mutations in HCMV. *Bioinformation*. 2011; 5(9):390. **(Impact Factor: 1.9)**
14. Gandra M, Umashankar V, Kasinathan N, Krishnan T, **Ravichandran S** et al. Molecular screening of CYP4V2 gene in Bietti crystalline dystrophy. *Molecular Vision*. 2011; 17:1970. **(Impact Factor 1.4)**

In Press

15. Grabauskas T, Trinity L, Verschoor CP, Marches R, Thibodeau A, Nehar-Belaid D, Eryilmaz G, Mahajan AS, **Ravichandran S** et al. CMV-specific clonal expansion of Th1, GZMK+ CD8+, and TEMRA T cells revealed by human PBMC single-cell profiling. *Genome Biology*. In press, 2025. **(Impact Factor: 9.1)**

Preprints

16. **Ravichandran S**, Bhatt B, Shah A, Das D, Balaji KN, Chandra N. Knock-down of a regulatory barcode shifts macrophage polarization from M1 to M2 and increases pathogen burden upon *S. aureus* infection. *eLife* (reviewed preprint).

Conference Proceedings

17. **Ravichandran S**, Ucar D, Marches R, Chaussabel D, Kuchel GA, Banchereau J. Variation in the single-cell epigenomic and transcriptomic profiles in response to seasonal influenza vaccination: a comparison between healthy young and older adults. *The Journal of Immunology*. 2024; 212(1_Suppl). **(Impact Factor: 3.3)**
18. **Ravichandran S**, Erra-Diaz F, Karakaslar OE, Marches R, Rossi R, Nahm MH, Chaussabel D, Kuchel GA, Banchereau J, Ucar D. Pre-vaccination CD56dim CD16+ NK cell abundance and Th1/Th17 ratio predict responsiveness to conjugated pneumococcal vaccine in older adults. *The Journal of Immunology*. 2023; 210(Suppl_1). **(Impact Factor: 3.3)**
19. Ucar D, Kuchel GA, Banchereau J, **Ravichandran S**, Karakaslar OE, Erra-Diaz F, Marquez E, Marches R. Sexual Dimorphism in Human Immune System Aging. *Innovation in Aging*. 2022; 6(Suppl_1):165. **(Impact Fctor 4.3)**

PATENTS

1. **Ravichandran S**, Ucar D, Banchereau J. Methods for Assessing Pneumococcal Vaccine Responsiveness. U.S. Provisional Patent Application No. 63/741,702. Filed Jan 3, 2025. **(Pending)**
2. **Ravichandran S**, Chandra N. Biomarkers and Associated Methods for Detecting Bacterial and Viral Infections in a Blood Sample. Indian Patent No. 404211. Filed Apr 10, 2020; **Granted** Aug 23, 2022.

INVITED TALKS & PRESENTATIONS

Invited Talks

1. Genomic Signatures of Acute Infections and Vaccine Responsiveness. Ramaiah University of Applied Sciences, Bangalore, India. December 2, 2025.
2. Distinct Baseline Immune Characteristics Associated with Responses to Conjugated and Unconjugated Pneumococcal Polysaccharide Vaccines in Older Adults. 8th Northeast Glenn Symposium on the Biology of Aging, Yale University. Oct 2024. **(Podium Presentation)**
3. A single-cell transcriptome atlas of Fluzone responses in healthy young and older adults. Systems Immunology in Aging and Complex Diseases, The Jackson Laboratory, Farmington, CT. June 2024. **(Invited Talk; Co-Organizer)**

4. Distinct Immune Characteristics Associated with Vaccine Responsiveness in Older Adults. AAI Annual Meeting – IMMUNOLOGY 2023, Washington, DC. (**Podium & Poster; AAI Trainee Abstract Award 2023**)
5. Identification of a Robust Blood-Based Biomarker Signature for Chronic Systemic Inflammation. Keystone Symposia – Translational Systems Immunology, Snowbird, UT, 2018. (**Podium & Poster; Global Travel Award**)

Selected Poster Presentations

1. Activation of cDC2 and monoDCs on Day 1 is associated with stronger responses to Fluzone HD in older adults. 9th Northeast Glenn Foundation Symposium, University of Connecticut, 2025.
2. Distinct baseline immune characteristics associated with responses to conjugated and unconjugated pneumococcal vaccines in older adults. American Geriatrics Society (AGS) Annual Meeting. 2024. (**Presidential Poster Session**)
3. Pre-vaccination CD56dim CD16+ NK cells and Th1/Th17 ratio predict responsiveness to conjugated pneumococcal vaccine in older individuals. 32nd Annual Short Course on Experimental Models of Human Cancer, The Jackson Laboratory, Bar Harbor, ME, 2023. (**Best Poster Award**)
4. Pre-vaccination CD56dim CD16+ NK cells and Th1/Th17 ratio predict responsiveness to conjugated pneumococcal vaccine in older individuals. 2nd Biennial Meeting, Sex Differences in the Immune System, Trinity College Dublin & Whitehead Institute, 2022 (**Best Poster Award**)

PROFESSIONAL & SCIENTIFIC SERVICE

Editorial & Peer Review Service

- **Research Topic Editor & Handling Editor**

1. *Frontiers in Aging* (IF: 4.3) - “Enhancing Vaccine Responsiveness in the Aging Population Through Advanced Immunology” (3,686 views)

- **Associate Editor / Handling Editor**

1. *Frontiers in Public Health* (IF: 3.4) - 4 manuscripts
2. *Frontiers in Immunology* (IF: 5.9) - 1 manuscript
3. *Frontiers in Aging* (IF: 4.3) - 1 manuscript

- **Peer Reviewer: 18 manuscripts (23 review instances, including multiple rounds) across 11 international journals and 3 publishers**

1. *npj Digital Medicine* (IF: 15.2) - 2 reviews
2. *npj Vaccines* (IF: 6.5) - 1 review
3. *Frontiers in Immunology* (IF: 5.9) - 2 reviews
4. *Frontiers in Endocrinology* (IF: 4.6) - 2 reviews
5. *MDPI Cancers* (IF: 4.4) - 2 reviews
6. *Frontiers in Aging* (IF: 4.3) - 1 review
7. *MDPI Current Oncology* (IF: 3.4) - 2 reviews
8. *MDPI Diagnostics* (IF: 3.3) - 4 reviews
9. *Frontiers in Medicine* (IF: 3.0) - 2 reviews
10. *Frontiers in Genetics* (IF: 2.8) - 3 reviews
11. *BMC Infectious Diseases* - 1 review

Conference Leadership

- **Co-Organizer & Abstract Evaluator:** Systems Immunology in Aging and Complex Diseases Symposium, The Jackson Laboratory, Farmington, CT. NIH Grant 5R13AG069519 (2024)
- **Session Moderator:** Systems Immunology in Aging and Complex Diseases Symposium, The Jackson Laboratory (2022, 2023)

Professional Memberships & Leadership

- American Association of Immunologists (Associate Member)
- American Geriatrics Society (Member)
- Vice President (2023–2024), Women in Science and Engineering (WiSE) Program, The Jackson Laboratory for Genomic Medicine (Internal programming)

MEDIA COVERAGE

2026

- **JAX News:** "Why do vaccines work for some, but not others?" - <https://www.jax.org/news-and-insights/2026/january/why-do-vaccines-work-for-some-but-not-others->

2024

- **Medical Xpress [Science X network, 10M+ monthly readers]:** "Researchers investigate why vaccines don't work as well in some older adults." -<https://medicalxpress.com/news/2024-01-vaccines-dont-older-adults.html>
- **Mirage News [independent real-time media portal, 1,500+ source entities]:** "Jackson Lab, UConn Health probe vaccine efficacy in older adults." - <https://www.miragenews.com/jackson-lab-uconn-health-probe-vaccine-efficacy-1152108/>
- **Technology Networks [10M+ annual visitors]:** "Why Do Vaccines Not Work Well in Some Older Adults?" -<https://www.technologynetworks.com/biopharma/news/why-do-vaccines-not-work-well-in-some-older-adults-382530>

2022

- **JAX News:** "Brooks Scholar Award — Bridging Cancer and Aging Research" - <https://www.jax.org/news-and-insights/2022/December/brooks-scholar-award-2022>

2021

- **The Hindu [5.3 million readers per day]:** "New blood-based biomarker to distinguish between bacterial and viral infections." - <https://www.thehindu.com/news/national/karnataka/new-blood-based-biomarker-to-distinguish-between-bacterial-and-viral-infections/article34577658.ece>
- **Times of India [13.5 million readers per day]:** "IISc team's biomarkers can spot bacterial, viral infections." - <https://timesofindia.indiatimes.com/india/iisc-teams-biomarkers-can-spot-bacterial-viral-infections/articleshow/82728128.cms>
- **Medical Xpress [Science X network, 10M+ monthly readers]:** "New blood-based biomarker to distinguish between bacterial and viral infections." - <https://medicalxpress.com/news/2021-05-blood-based-biomarker-distinguish-bacterial-viral.html>
- **BioSpectrum India [550,000+ monthly readers]:** "IISc identifies biomarkers to differentiate between bacterial, viral infections." - <https://www.biospectrumindia.com/news/58/18735/iisc-identifies-biomarkers-to-differentiate-between-bacterial-viral-infections.html>

- **Vigyan Prasar / India Science and Technology (pg No: 7)[Government of India, DST]:** Featured in COVID newsletter, Oct 2021 - <https://www.indiascienceandtechnology.gov.in/images/covidnewsletter-26-10-2021.pdf>

2019

- **The Hindu [5.3 million readers per day]:** "IISc researchers develop protein to kill drug-resistant bacterium." -<https://www.thehindu.com/news/cities/bangalore/iisc-researchers-develop-protein-to-kill-drug-resistant-bacterium/article28751864.ece>
- **Phys.org[Science X network, 10M+ monthly readers]:** "A novel peptide to rapidly kill multidrug-resistant bacteria." - <https://phys.org/news/2019-07-peptide-rapidly-multidrug-resistant-bacteria.html>

TECHNICAL SKILLS

- **Immunology & Translational Research:** Human vaccine cohort studies, antibody response profiling, innate and adaptive immune analysis, flow cytometry, sex-specific immune profiling
- **Multi-omic Profiling:** scRNA-seq, snATAC-seq, bulk RNA-seq, NanoString, Olink, NULISaseq, microarray (Agilent, Affymetrix, Illumina)
- **Computational & ML:** WGCNA, ARACNe, GSEA, Elastic Net, LASSO, Random Forest; Seurat, Scanpy, ArchR, Bioconductor; R, Python, Bash, HPC/SLURM